# How close does a fabricated silicon waveguide come to its simulation? Seven Mach-Zehnder interferometers to measure the group index

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**Abstract—How close does a fabricated silicon waveguide come to its simulated group index? I propose seven unbalanced Mach-Zehnder interferometers (MZIs) to find out. In each MZI, light from a short and a long arm adds together, and the output rises and falls with wavelength. The spacing of the peaks, the free spectral range (FSR), gives the group index. The arms differ by 30 to 200 µm. A compact model fitted to Lumerical MODE predicts a group index of 4.19 at 1550 nm and FSRs from 19.1 nm to 2.9 nm. A grating-coupler loopback on the same chip records the couplers alone, so their shape can be removed from every MZI spectrum.**

## I. Introduction

How fast does light travel in a silicon waveguide? For a pulse, the group index $n_g$ sets the answer: the pulse moves at $c/n_g$ [1, Eq. (3.4)]. The same number sets how far apart the resonances of every ring and interferometer on a chip lie [1, p.65]. A designer can calculate $n_g$, but a fabricated waveguide never matches its drawing. The top silicon layer alone varies in thickness by about ±5 nm [1, p.49], and the etched width varies too [1, Sec. 11.1]. The value that matters is the one on the chip.

An unbalanced Mach-Zehnder interferometer (MZI) measures that value. A Y-branch splits the light into two arms, one longer than the other by ΔL, and a second Y-branch adds the two back together. At some wavelengths the two waves arrive in step and add; at others they arrive out of step and cancel. The output therefore rises and falls with wavelength, and the spacing of its peaks depends on $n_g$ and ΔL. This design places seven MZIs on one chip, with ΔL from 30 to 200 µm, to measure $n_g$ and compare it with the simulation.

## II. Theory

For two ideal 50/50 Y-branches and arms of length $L_1$ and $L_2$, the transmission is [1, Eq. (4.18)]

$$T(\lambda) = \tfrac{1}{4} \cdot \left| e^{-i\beta L_1} + e^{-i\beta L_2} \right|^2, \quad \beta = 2\pi\, n_{eff}(\lambda)/\lambda - i\alpha/2 \qquad (1)$$

where $n_{eff}$ is the effective index and α the propagation loss. Without loss, (1) becomes $T = \tfrac{1}{2}[1 + \cos(\beta\Delta L)]$, with $\Delta L = L_2 - L_1$ [1, Eq. (4.19c)].

In circuit terms, the MZI is a feed-forward comb filter: a signal added to a copy of itself that arrives τ later. The peaks of such a filter lie $1/\tau$ apart in frequency. In the MZI, τ is the extra time light spends in the long arm. That time is a group delay, $\tau = n_g \Delta L/c$, so the peaks lie $c/(n_g \Delta L)$ apart in frequency, or, in wavelength [1, Eq. (4.20b)],

$$FSR = \lambda^2 / (n_g \cdot \Delta L), \quad n_g = n_{eff} - \lambda \cdot dn_{eff}/d\lambda \qquad (2)$$

The second part of (2) defines the group index [1, Eq. (3.5)]. The effective index fixes where each peak sits; the group index fixes how far apart the peaks are. For example, $\Delta L = 100$ µm and $n_g = 4.19$ give peaks 5.73 nm apart at 1550 nm. Solved for $n_g$, (2) turns a measured FSR into a group index:

$$n_g = \lambda^2 / (FSR \cdot \Delta L) \qquad (3)$$

Here λ is the wavelength midway between the two peaks that define the FSR.

## III. Modelling and simulation

### A. Waveguide

The waveguide is a silicon strip, 500 nm wide and 220 nm high, in oxide. It carries the fundamental quasi-TE mode at 1550 nm. Fig. 1 shows that mode: most of the power flows inside the silicon, and the field jumps at the side walls, where the index drops from 3.48 to 1.44. The finite-difference eigenmode solver of Lumerical MODE (Palik materials, a 2.5 × 1.7 µm region, a 10 nm mesh) gives $n_{eff} = 2.445$. An independent finite-element solver, femwell [5], agrees within 0.001.

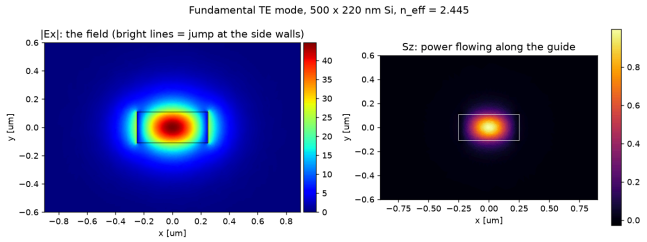



Fig. 1. Fundamental TE mode of the 500 × 220 nm strip at 1550 nm (femwell): $|E_x|$ (left) and the power flow $S_z$ (right). $n_{eff} = 2.445$.

### B. Compact model

One wavelength is not enough, because the MZIs are measured from 1500 to 1600 nm. A wavelength sweep in MODE, fitted with a quadratic about $\lambda_0 = 1.55$ µm, gives the compact model (λ in µm):

$$n_{eff}(\lambda) = 2.4446 - 1.1265\,(\lambda - 1.55) - 0.0336\,(\lambda - 1.55)^2 \qquad (4)$$

From (2), $n_g = n_1 - n_2\lambda_0 = 2.4446 + 1.1265 \times 1.55 = 4.191$ at 1550 nm. Fig. 2 shows why $n_g$ is so much larger than $n_{eff}$. The effective index falls as the wavelength grows, so the term $-\lambda \cdot dn_{eff}/d\lambda$ adds 1.75. Two effects make it fall: silicon's own index drops with wavelength, and a longer wavelength pushes more of the mode out of the silicon into the oxide [1, p.66].

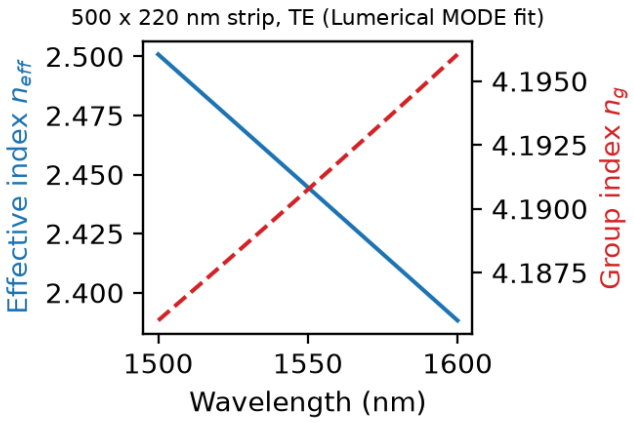



Fig. 2. Effective index (solid) and group index (dashed) of the strip waveguide from the compact model (4).

### C. Interferometer design

Each MZI uses two Y-branches (ebeam_y_1550 [2], [3]) and two grating couplers (GC_TE_1550_8degOxide_BB) 127 µm apart, the pitch of the fibre array. The short arm is 125.8 µm in every MZI; only the long arm grows. Two limits set the range of ΔL. The grating couplers pass light over about 50 nm, so the FSR must stay well below 50 nm for several peaks to appear. A long ΔL costs chip area and makes the peaks crowd together; even so, the smallest FSR here, 2.9 nm, still spans 2,900 steps of the 1 pm laser. ΔL from 30 to 200 µm keeps every FSR between 2.9 and 19.1 nm (Table I).

The couplers shape every spectrum, so the chip also carries a loopback: two grating couplers joined by one waveguide. It records the couplers alone. Dividing each MZI spectrum by it removes their shape (de-embedding).

TABLE I. DESIGN VARIATIONS (TE, 1550 NM)

| Device | ΔL (µm) | $L_2$ (µm) | FSR, (2) (nm) | FSR, simulated (nm) | $n_g$ from (3) |
|---|---|---|---|---|---|
| MZI1 | 30 | 155.8 | 19.11 | 19.03 | 4.1906 |
| MZI5 | 40 | 165.8 | 14.33 | 14.23 | 4.1902 |
| MZI2 | 50 | 175.8 | 11.47 | 11.53 | 4.1912 |
| MZI6 | 75 | 200.8 | 7.64 | 7.63 | 4.1908 |
| MZI3 | 100 | 225.8 | 5.73 | 5.74 | 4.1909 |
| MZI7 | 150 | 275.8 | 3.82 | 3.82 | 4.1912 |
| MZI4 | 200 | 325.8 | 2.87 | 2.87 | 4.1904 |

### D. Transmission spectra

Fig. 3 shows what the measurement should look like. It uses (1) with the compact model (4), a loss of 3 dB/cm ($\alpha = 6.91 \times 10^{-5}$ µm$^{-1}$), the arm lengths of the layout, and the S-parameters of the library grating coupler [3]. In grey, each MZI alone swings between full transmission and deep nulls, and the swings crowd together as ΔL grows: about 5 peaks across 100 nm in MZI1, about 35 in MZI4. In blue, the two grating couplers add their own shape, 4.7 dB of loss at best near 1554 nm and falling away on both sides. The loopback (green) is that shape alone, which is why it is on the chip.

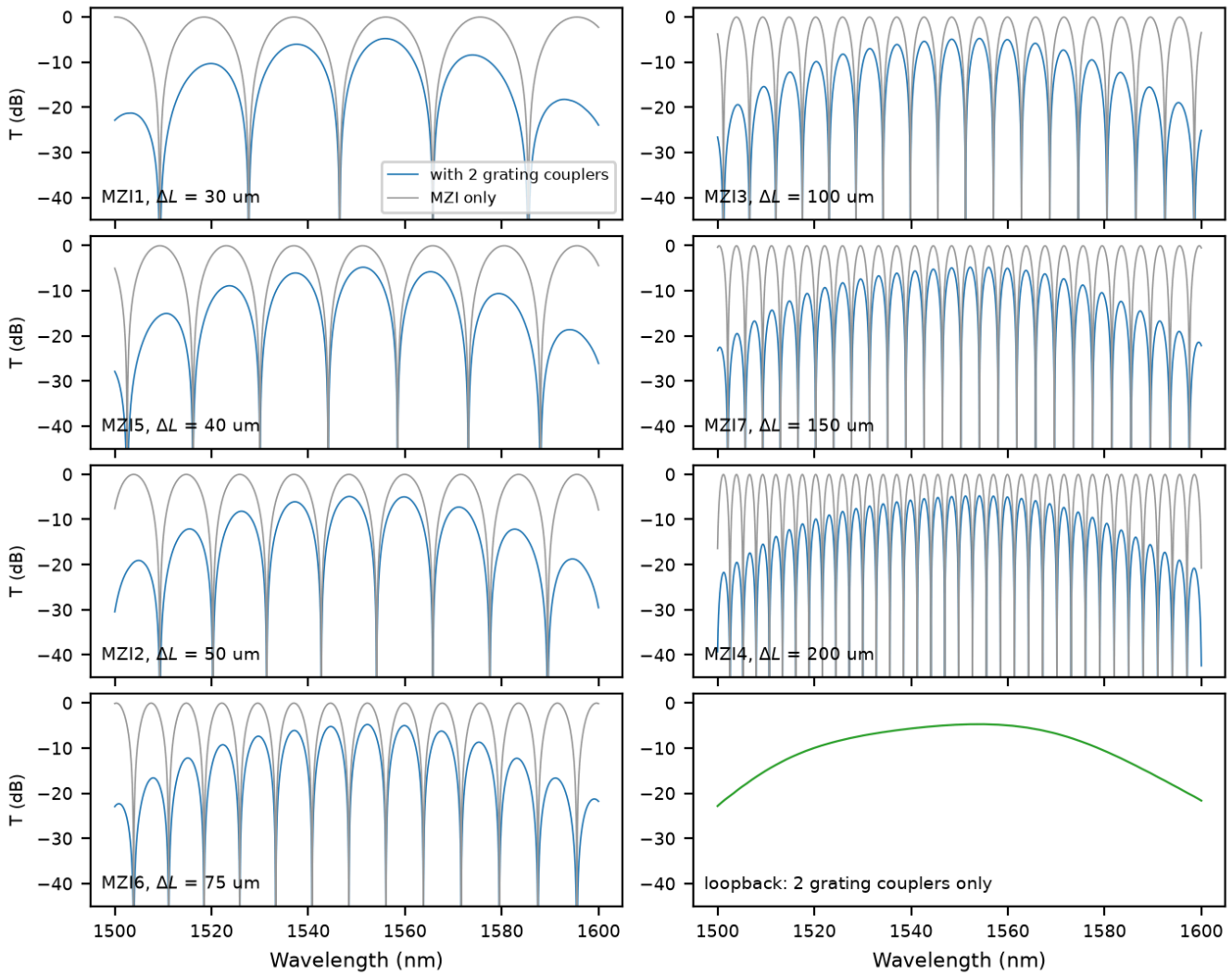



Fig. 3. Simulated transmission of the seven MZIs: the MZI alone (grey) and with two grating couplers (blue); last panel: the loopback (two grating couplers only).

### *E. Testing the extraction*

Does (3) give back the $n_g$ that went into the model? I found the two peaks nearest 1550 nm in each simulated spectrum, sampled in 1 pm steps like the laser, took their spacing as the FSR, and applied (3). All seven results lie between 4.1902 and 4.1912 (Table I), within 0.001 of the model's 4.1908. With measured spectra I will divide by the loopback, fit the peak positions rather than read them, and check that the seven MZIs agree on one $n_g$.

## IV. Fabrication

The layout (Fig. 4) uses the SiEPIC EBeam PDK [3] in KLayout and fits the 605 × 410 µm cell in two rows of four devices. It passes the SiEPIC layout verification and the repository's checks with no errors, and is merged into the openEBL-2026-10 layout [4]. Each device has an automated-test label on its upper grating coupler, the laser input (opt_in_TE_1550_device_Diako1_MZI1 to MZI7, and _loopback). *Fabrication details: to be completed after fabrication.*

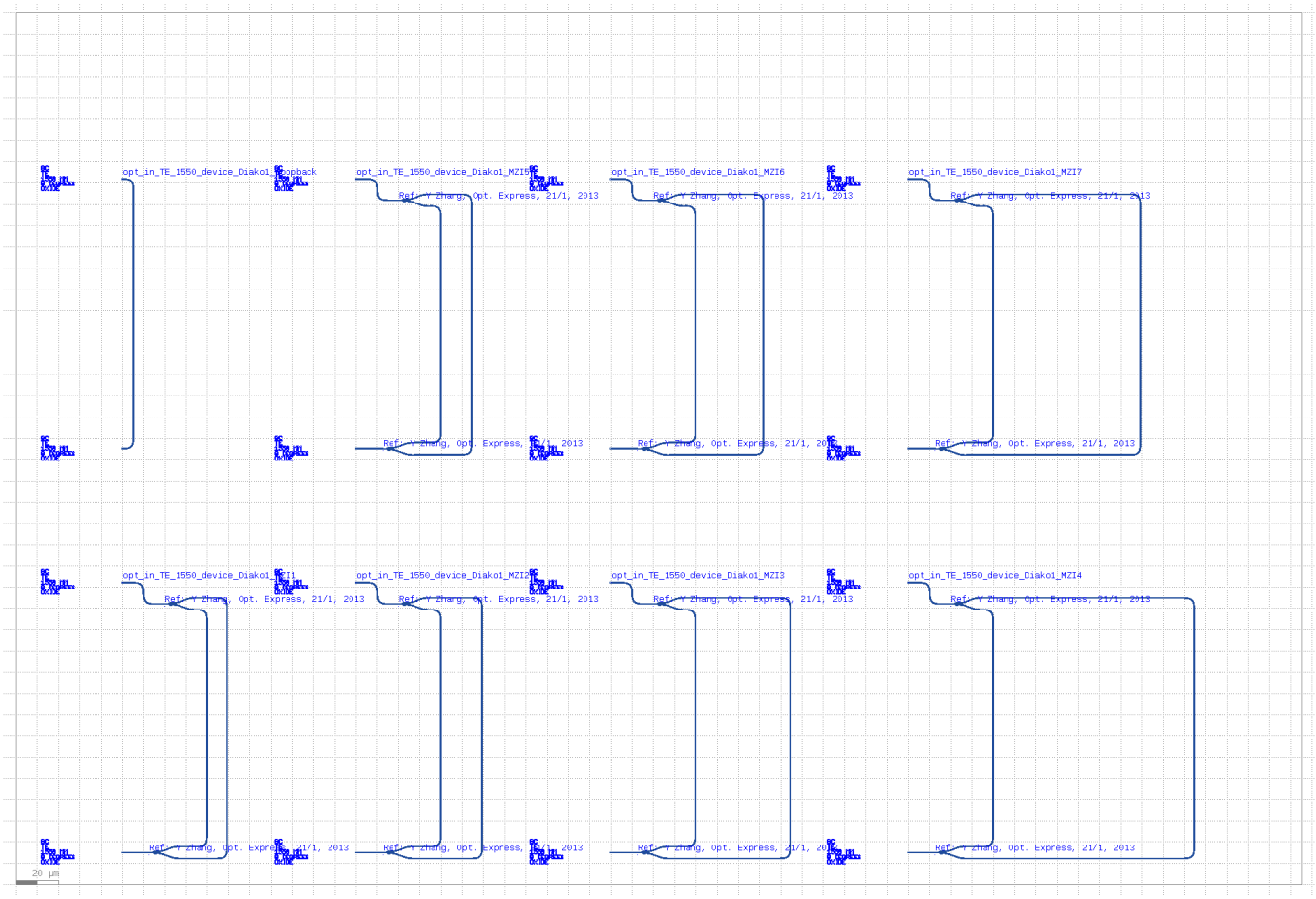



Fig. 4. Final layout EBeam_Diako1. Bottom row: MZI1–MZI4 (ΔL = 30, 50, 100, 200 µm). Top row: the loopback and MZI5–MZI7 (ΔL = 40, 75, 150 µm). Each long arm makes a wider loop.

## V. Experimental data

*To be completed after measurement.*

## VI. Analysis

*To be completed after measurement.*

## VII. Conclusion

*To be completed after measurement.*

## References


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